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Intelligent Maritime Surveillance Framework Driven by Fusion of Camera-based Vessel Detection and AIS Data

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***Abstract*— Efficient maritime surveillance is necessary for navigation, which usually uses cameras to capture the vessels. However, the information in camera-based data is limited. In this work, we propose a novel intelligent maritime surveillance framework driven by the fusion of camera-based vessel detection and Automatic Identification System (AIS) data. Firstly, we employ a vessel detection network to get the relative positions of the vessels from the calibrated camera-based data. Meanwhile, we design a series of filters based on data completeness, detection range, and vessel course to exclude the invalid AIS data. In the end, we propose a data fusion module based on estimating the time when the vessel arrive at the specific position. According to the experiment on our collected dataset, the proposed framework performs competitively in diversified scenes. The mean absolute distance deviation of the estimation is less than 30 meters, and the accuracy of data fusion is 81.423%.**

I. Introduction

With the rapid development of artificial intelligence, many researchers employ intelligent methods in maritime tasks like auxiliary navigation, illegal activity identification, and, primarily, maritime surveillance. After the widespread employment of imaging devices, camera-based data analysis became the dominant way to sense the navigation environment. However, the expressed information of camera-based data is insufficient. To tackle this problem, we propose an intelligent maritime surveillance framework driven by the fusion of camera-based vessel detection and AIS data.

For the camera-based data analysis, the first step is calibration, which is set to correct the deviation caused by camera distortion. The most widely used method is proposed by Zhang *et al.* [1], which only requires the camera to capture a planar pattern at different orientations. After the calibration, the output data is used for vessel detection, including traditional mathematical model-based and deep learning-based methods.

In inchoate vessel detection tasks, researchers used some mathematical model-based methods like Histogram of Oriented Gradients (HOG) feature descriptor [2] and Deformable Part Model (DPM). HOG improved the scale-invariant feature transform and shape contexts, which was also employed in maritime surveillance. For instance, Dong *et al.* [3] proposed a multi-level ship detection algorithm based on HOG and Shi *et al.* [4] used HOG to valid the vessel detection results. DPM regarded detection tasks as the philosophy of ”divide and conquer”, which archived competitive performance after years of development [5], [6], [7]. To improve the speed and accuracy of training, DPM firstly proposed some important innovations like ”hard negative mining ” and ”bounding box regression”, which are still used in recent methods. For instance, Li *et al.* [8] employed DPM to identify the vessels, Song *et al.* [9] proposed a vessel detection method based on DPM.

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Fig. 1. The example of comparison between the normal camera-based vessel detection and the output of the proposed framework.

In 2012, AlexNet [10] won the ImageNet image classification competition, which showed the striking potential of convolutional neural network (CNN) in computer vision tasks. From then on, deep learning are widely employed in image processing tasks like image classification, image enhancement and primarily, object detection, which generally includes two stages-methods and one stage-methods [11]. In 2014, the Regions with CNN features (RCNN) [12] was firstly employed in object detection tasks. RCNN is a typical two stages-method, which firstly uses selective search to extract object proposals, then rescale and feed them into CNN, which includes linear SVM classifiers in the last part to get the prediction. Following RCNN, many methods were proposed like Fast RCNN [13], Faster RCNN [14], which archive significant performance in object detection tasks. Due to their competitive performance, many researchers tried to employ them in maritime surveillance, e.g. Kang *et al.* [15] proposed a modified faster RCNN model to detect vessels. However, two stages-methods still suffer from low computation speed compared with one stage-methods.

Unlike the two stages-methods, one stage-methods try to detect the object with a one period neural network. Yolo [16] was firstly proposed in 2015, which divide each image into several patches. Each patch needs to predict bounding boxes and probabilities simultaneously. In the following years, Yolo

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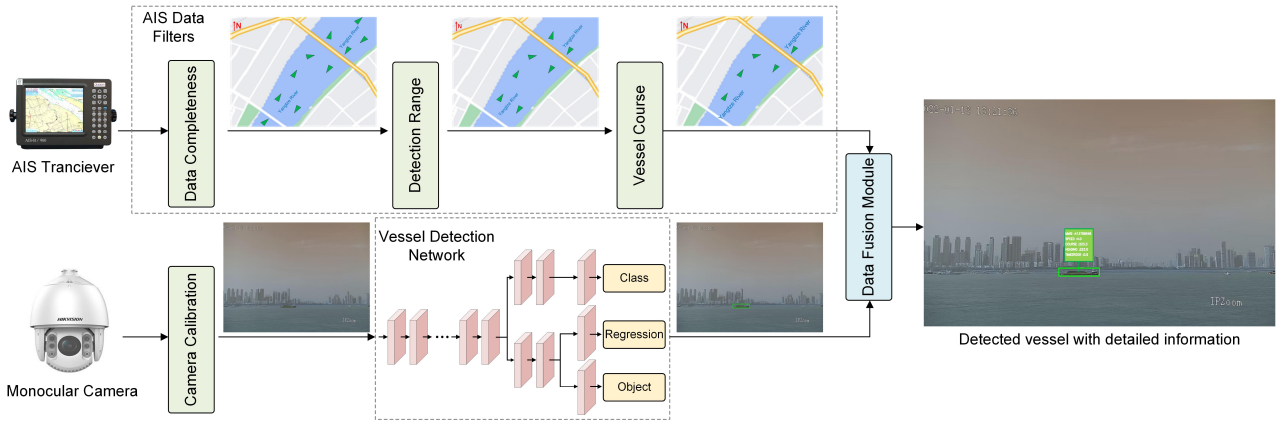

Fig. 2. The flowchart of the proposed framework. Firstly, the AIS data is received and fed into AIS data filters. Meanwhile, the camera-based data is captured by a calibrated monocular camera and analyzed by the vessel detection network to get the vessels' positions. The data fusion module finally fuses them to provide the intuitive data for maritime surveillance.

has been updated to many versions [17], [18], [19], which are also widely employed in maritime tasks, e.g., Liu *et al.* [20] proposed an enhanced Yolov3 network for vessel detection in different weather conditions and Jiang *et al.* [21] employed Yolov4 as the basic vessel detection model, which archived competitive performance in the experiment.

Recently, Megvii Technology proposed an object detection method named Yolox [22], which achieves better results in both efficiency and accuracy compared with past versions of Yolo. Considering the practical scenes our proposed framework work in, we build a Yolox-based vessel detection network, which can detect vessels accurately and timely. Besides, to comprehend more information about the detected vessels, we fuse the messages obtained from AIS into the camera-based data.

AIS is a vessel navigation assistant system, in which the basic information of each vessel, including the Maritime Mobile Service Identify (MMSI), course, position, and speed, is transferred on Very High Frequency (VHF) band [23]. Fusion with AIS makes camera-based data express richer information, as shown in Fig. 1. Before our work, Chen *et al.*[24] combined the video and AIS data, which is helpful for specific vessel tracking in maritime video surveillance. However, it is powerless in multiple vessels scenes. Lu *et al.* [25] proposed a fusion algorithm based on the estimation of the distance and vessels' length, which is restricted by relative azimuth angle between the course of the vessel and the camera's bearing. To avoid the limitation, we propose a fusion method based on estimating the time when the vessel arrives at the target position. According to the experiment, our fusion method can accurately match the vessels in camera-based data with their AIS information respectively.

The main contributions of our work are as follows

- We propose a novel intelligent maritime surveillance framework. For camera-based data, we design a Yolox-based vessel detection network, which is able to obtain the vessel's position accurately.
- To exclude insufficient AIS data, we propose a series of AIS data filters based on the camera parameters and detailed information in AIS data.
- We design a data fusion module working with the target position, the horizontal visual field of view (FOV), and the most recent AIS data to estimate the time when the vessels arrive at the target positions.
- According to the experiment, our framework work sufficiently in diversified maritime scenes with competitive fusion accuracy.

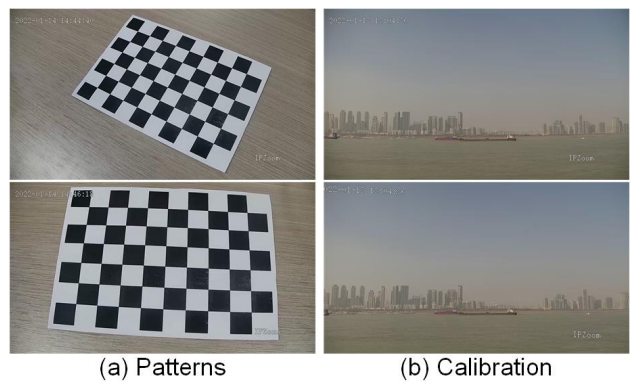

Fig. 3. (a) Examples of the patterns with 10 × 7 squares in camera calibration. (b) Comparison between the original and calibrated images.

II. Proposed Method

A. Camera Calibration

In the proposed framework, the relative position of the vessel is an essential parameter, which is greatly affected by image accuracy. However, with the long-time preservation, some precision components of the camera were deformed, which leads to distortion in generated images. To reduce the errors caused by deformation, the camera-based data should be calibrated before the dataset collection. We choose Zhang's method [1] and collect 20 images of the pattern with 10 × 7 squares for the calibration, as shown in Fig. 3 (a). With the calculation of the calibration metrics, the captured images will be transformed to the actual scale, as shown in

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Fig. 4. (a) The example of the detection results. (b) The sketch map of the detection range filter. (c) The sketch map of the vessel course filter.

Fig. 3 (b). Then the calibrated images are fed into the vessel detection network.

B. Vessel Detection Network

To get the vessels' positions in camera-based data, we design a Yolox-based vessel detection network. Due to the further information like the categories will be extracted from the matched AIS data, we set the number of output classes to one and optimize the loss function to make the network only concentrate on position extraction. In the end, we calculate the center coordinate of the vessel (X_v, Y_v) as the relative position in the image, as shown in Fig. 4 (a).

C. AIS Data Filters

The sending period of AIS data is set differently among the diverse vessels. Besides, when the course and speed of the vessel have considerable fluctuation, the vessel-end AIS transceiver will publish the AIS data more frequently. Therefore, the time resolution of AIS data is poor, which can only be distinguished by the local time of the receiving device. Besides, unlike reliable transmission protocols, the AIS transceiver is only responsible for publishing data to the VHF bandand is not concerned about the destination and integrity of the data transmission. Therefore, some invalid data are collected in the dataset, such as the fragmentary data and the data outside the surveillance range. In order to avoid the interference of these data on the fusion results and reduce the computational complexity, we propose a series of AIS data filters based on data completeness, detection range, and vessel course.

1) Data Completeness Filter: Due to the unreliable transmission, much of the AIS data we receive is mutilated. In the decoded AIS data, the data loss manifests as lack, deviation, and garbled code. Therefore, we reject AIS data with critical

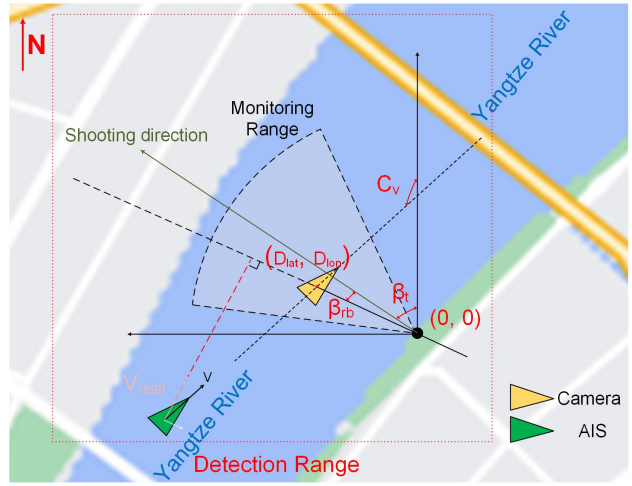

Fig. 5. The sketch map of the arrival time estimation. The green vessel represents the recent positions in AIS data the yellow vessel means the detected vessel in camera-based data.

data missing, abnormal values, and garbled data in the first filtering step. After data completeness filter $\mathcal{F}_{dc}$, the output AIS data D_{dc} can be expressed as

$$D_{dc} = \mathcal{F}_{dc}(D_{all}, K, M, T), \tag{1}$$

where D_{all} is all of the collected AIS data, T represents the time span, which is set to 120 seconds in the experiment, and K, M mean the necessary elements and their tolerant error ranges in the AIS data, respectively.

2) Detection Range Filter: For the long transmission distance, the shore-based devices are able to receive the AIS data extensively. However, in our research, we only focus on the AIS data of the vessels that are not far from the surveillance area, as shown in Fig. 4 (b), so we design a detection range filter based on the location of the camera. The output AIS data D_{dr} after range filtering $\mathcal{F}_{dr}$ can be expressed as

$$D_{dr} = \mathcal{F}_{dr}(D_{dc}, P_{camera}, \epsilon_{lon}, \epsilon_{lat}), \tag{2}$$

where P_{camera} means the position of the camera, ϵ_{lon} and ϵ_{lat} represent the restricted range in longitude and latitude, which is set to 0.1 in our framework.

3) Vessel Course Filter: The courses and speeds of the vessels can be extracted from the AIS data. Besides, based on the horizontal FOV angle and the location of the camera, we estimate the monitoring range as a fan-shaped field. Therefore, the vessel course filter is proposed to extract the record of vessels sailing or will sail into the monitoring range, as shown in Fig. 4 (c). The output data D_{vc} of the vessel course filter $\mathcal{F}_{vc}$ can be expressed as

$$D_{vc} = \mathcal{F}_{vc}(D_{dr}, P_{camera}, O_{camera}, \beta_{FOV}), \tag{3}$$

where P_{camera}, O_{camera} and β_{FOV} represent the location, shooting orientation, and horizontal FOV angle of the camera, respectively.

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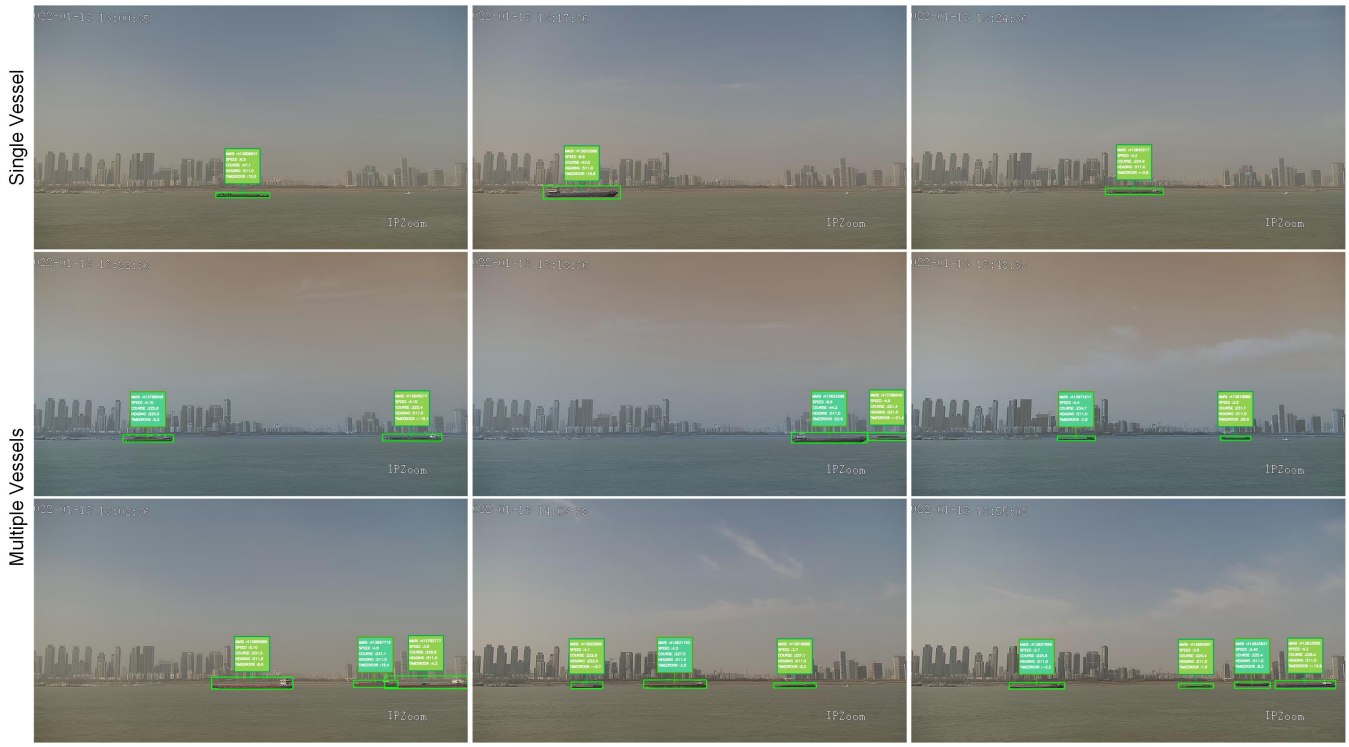

Fig. 6. The performance of the proposed data fusion module in different scenes (Enlarge the figure to see more details).

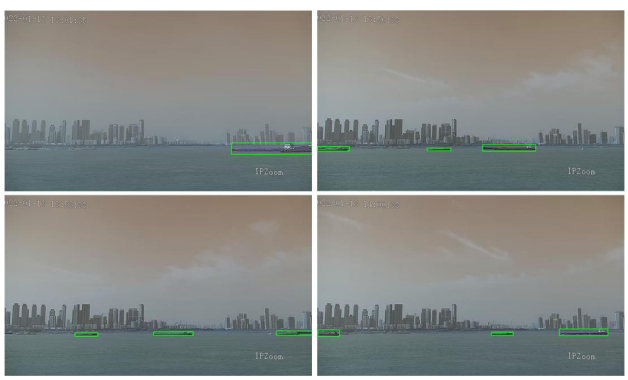

Fig. 7. Examples of the inaccurate detection results, which always happens in multiple vessels scenes.

D. Data Fusion Module

The vessel detection network provide the vessels position coordinates in the camera-based data, and corresponding valid recent AIS data is extracted from AIS data filters. We then propose a data fusion module based on the arrival time estimation. In the first place, we convert the latitude and longitude differences to absolute distances. In the northern hemisphere, the distance D between $A(LonA, LatA)$ and $B(LonB, LatB)$ can be calculated by

$$D = \frac{\pi R \cdot \arccos C}{180}, \tag{4}$$

where R is the radius of the earth and C can be expressed as

$$C = \cos LatA \cdot \cos LatB \cdot \cos(LonA - LonB) + \sin LatA \cdot \sin LatB. \tag{5}$$

In addition, a straight line $\mathcal{L}$ can be formed between the detected vessel and the camera, which can be expressed as

$$tan(\beta_t + \beta_{rb})X - Y = 0, \tag{6}$$

where β_{rb} is calculated as follows

$$\beta_{rb} = (0.5 - \frac{X_v}{W}) \cdot \beta_{FOV}, \tag{7}$$

where the directions of x-axis and y-axis are north and west respectively. β_t represents the shooting direction of the camera. β_{FOV} is the horizontal FOV angle of the camera. X_v means the horizontal coordinate of the camera-based vessel detection result, and W is the width of the camera-based data. As shown in Fig. 5, with the distance conversion, the time when the vessel pass $\mathcal{L}$ can be estimate by

$$T = \frac{|tan(\beta_t + \beta_{rb}) \cdot D_{lat} - D_{lon}|}{\sqrt{(tan^2(\beta_t + \beta_{rb}) + 1} \cdot V_{relat}}, \tag{8}$$

$$V_{relat} = \max[\mu \cos(90 - \beta_{rb} - C_v) \cdot V_{real}, \epsilon], \tag{9}$$

where D_{lat} and D_{lon} represent the north-south and east-west distances between the vessel and the camera. V_{relat} is the decomposition of the vessel's speed V_{real} in the direction perpendicular to $\mathcal{L}$. μ is the conversion factor of kilometers to nautical miles set to 1.852, C_v means the course of the vessel, and ϵ is set to 0.001, which is employed to avoid zero becoming the denominator.

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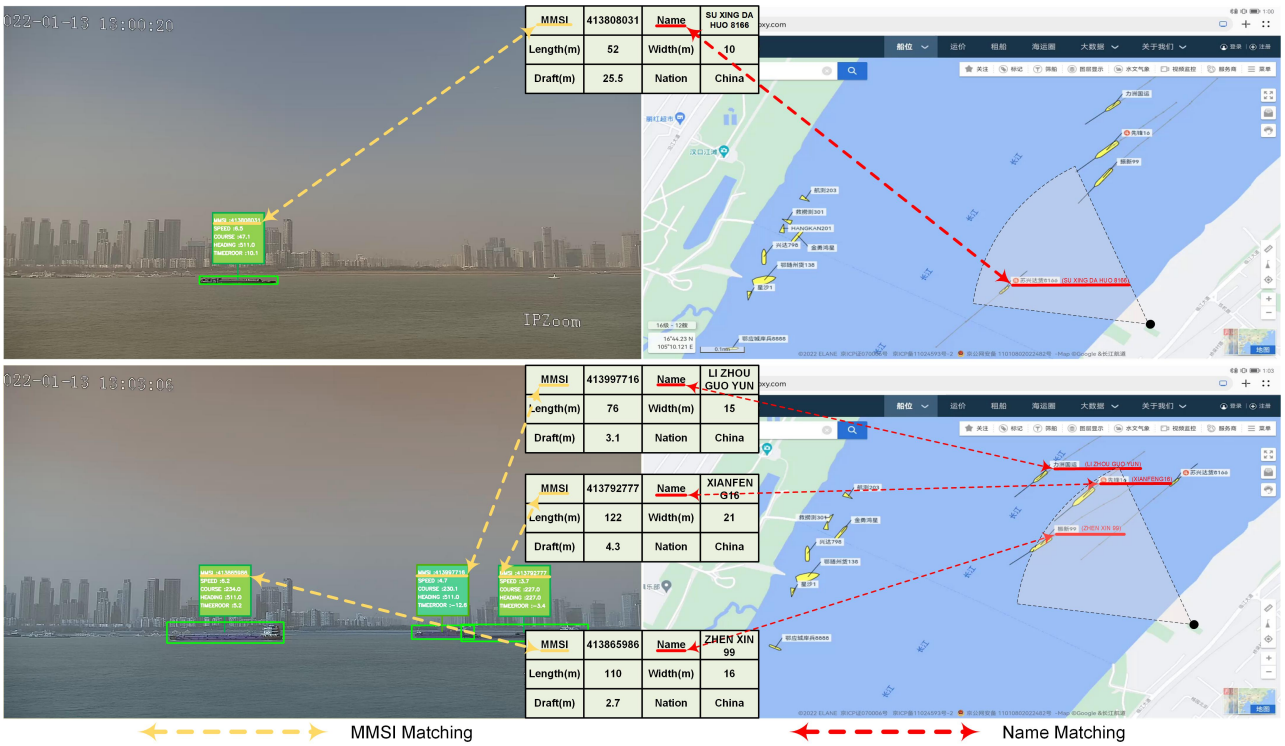

Fig. 8. Visual comparisons between the fusion result and the ground truth, where the vessels are labeled by MMSI and names separately. To match the detected targets with the corresponding vessels, we extract the AIS data from the AIS database as the criterion.

TABLE I
FUSION ACCURACY UNDER DIFFERENT SCENES

Metrics	Overall Performance	Single Vessel	Multiple Vessels
Accuracy	81.423%	91.803%	78.125%
Error	18.577%	8.197%	21.875%

TABLE II
MEAN ABSOLUTE ERROR OF CAMERA-BASED ARRIVAL TIME(UNIT: SECOND) ESTIMATION AND CORRESPONDING DISTANCE(UNIT: KM) IN DIFFERENT SCENES

Metrics	Overall Performance	Single Vessel	Multiple Vessels
Time	12.757	8.735	13.700
Distance	0.0286	0.0217	0.0300

III. Experiments

The dataset used in the experiment is collected with a monocular camera and an AIS receiver by the Yangtze River in Wuhan, China. However, in some special scenes, the proposed method will perform inaccurately. To our best knowledge, these scenes include the following.

1) Errors of AIS data: In most cases, every vessel must keep the AIS transceiver on during the navigation. However, especially in inland waterways, some vessels may illegally close AIS data transceivers or navigate with the damaged transceivers, which will publish inaccurate AIS data. In such conditions, the fusion module will work insufficiently.

2) Inaccurate Detection Results: To improve the detection accuracy, in the experimental scenes, a new dataset was collected for network training. However, some thorny problems still exist in detected results like target overlap and micro-target detection, as shown in Fig. 7. Due to the inaccurate detection, the data fusion module obtain wrong positions, which results in considerable deviations in the fusion process.

To evaluate the accuracy of the proposed framework, we randomly extract 200 frames consisting of 62 in single-vessel scenes and 138 in multiple-vessels scenes for testing. Besides, the inaccurate camera-based vessel detection result will be excluded to reduce the negative influence from the detection process. In the experiment, we evaluate the proposed framework with accuracy of fusion and deviation of arrival time and distance estimation.

A. Accuracy of Fusion

As shown in Fig. 7, the proposed method matches each detected vessel in images with an AIS record. For validation, we recorded the track of the vessels from the real-time maritime surveillance website [26] simultaneously as the ground truth, as shown in Fig. 8. The quantitative comparisons of different scenes are shown in Table. I. According to the experimental results, the framework is able to fuse the camera-based vessel detection result and AIS data accurately in single-vessel scenes, which is only influenced by wrong AIS data. In the multiple-vessels scenes, the device receives a larger amount of AIS data, which includes more inaccurate information. Besides, when the vessels are too close, the data fusion module is insufficient due to the deviations in arrival time estimation. Therefore, the accuracy of the fusion is lower in multiple-vessel scenes.

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B. Deviation of Arrival Time and Distance Estimation

To determine the scenes in which our framework has competitive performance, we regard the time when the images are captured as the truly arrival time to calculate the estimated error E_{Time}, which can be expressed as

$$E_{Time} = \frac{\sum_{N_{correct}}^{i} \left| T_{Est}^{i} - T_{gt}^{i} \right|}{N_{correct}}, \tag{10}$$

where $N_{correct}$ is the amount of the correctly fused vessels. T_{Est}^{i}, T_{gt}^{i} mean the estimated and real arrival time, respectively. With the error of the arrival time estimation, the average distance deviation between the estimated and captured positions can be calculated by

$$E_{Distance} = \frac{\sum_{N_{correct}}^{i} T^{i} \cdot V^{i}}{N_{correct}}, \tag{11}$$

where T^{i} means the mean absolute error of the time estimation, V^{i} is the speed of the vessel extracted from the AIS data. The results are shown in Tab. II, which indicates that our method is sufficient when the distance between the vessels is more than 30 meters.

IV. Conclusion

In this paper, we propose an intelligent maritime video surveillance framework driven by fusion of camera-based vessel detection and AIS data. Firstly, to get the relative positions of the vessels in camera-based data, we build a vessel detection network based on Yolox [22]. A series of AIS data filters are then employed to eliminate the insufficient AIS data. In the fusion process, we propose a module based on arrival time estimation to match and fuse the AIS data with the detected vessels.

According to the experimental results, the proposed framework is able to fuse the camera-based vessel detection results and AIS data with 81.423% accuracy generally, which has better performance in single-vessel scenes. Moreover, the calculation of distance deviation shows that our framework is sufficient when the distance between the vessels is more than 30 meters.

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